

Class AreaCalculator

java.lang.Object
AreaCalculator

public class **AreaCalculator**
extends **Object**

Purpose: Provides utility methods for geometric area calculations and an interactive interface for user calculations.

Constructor Summary

Constructors	
Constructor	Description
AreaCalculator()	

Method Summary

All Methods	Static Methods	Instance Methods	Concrete Methods
Modifier and Type	Method		Description
double	calculateRectangleArea (double length, double width)		Calculates the area of a rectangle based on its length and width.
double	calculateTriangleArea (double base, double height)		Calculates the area of a triangle based on its base and height.
static void	main(String [] args)		Main method to handle user interaction via the console.

Methods inherited from class java.lang.Object

clone , equals , finalize , getClass , hashCode , notify , notifyAll , toString , wait , wait , wait

Constructor Details

AreaCalculator

```
public AreaCalculator()
```

Method Details

calculateRectangleArea

```
public double calculateRectangleArea(double length,  
                                     double width)
```

Calculates the area of a rectangle based on its length and width.

Parameters:

length - The vertical or horizontal extent of the rectangle (double).

width - The horizontal or vertical extent of the rectangle (double).

Returns:

The total area as a product of length and width (double).

calculateTriangleArea

```
public double calculateTriangleArea(double base,  
                                    double height)
```

Calculates the area of a triangle based on its base and height.

Parameters:

base - The bottom side of the triangle (double).

height - The vertical height from the base to the opposite peak (double).

Returns:

The calculated area as $(\text{base} * \text{height}) / 2$ (double).

main

```
public static void main(String [] args)
```

Main method to handle user interaction via the console.

Parameters:

args - Command line arguments (not used).